Unit 4 Pre-Test

Sequences & Exponential Functions

Standards assessed: NC.M1.F-IF.3 · F-IF.7 · F-IF.8b · F-BF.1a · F-BF.2 · F-LE.1 · F-LE.3 · F-LE.5

Name: _____ **Date:** _____ **Period:** _____

SCORE *3.0* **Basic Understanding**

NC.M1.F-BF.2 · NC.M1.F-LE.1 · NC.M1.F-IF.7 Find patterns in arithmetic and geometric sequences.

• *I can find the next term, the common difference, and the common ratio, and tell linear from exponential.*

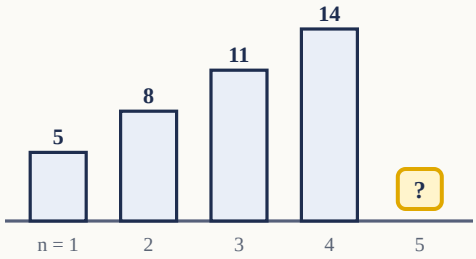
Answer each question. Use the picture to help you.

How to type your answers

number `-4` fraction `1/3` power `2^t` model `200(2)^t`

1. *Next term*

What is the next term? 5, 8, 11, 14, …

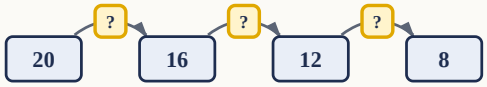


Answer: _____

answer: 17

2. *Common difference*

What is the common difference *d*? 20, 16, 12, 8, …

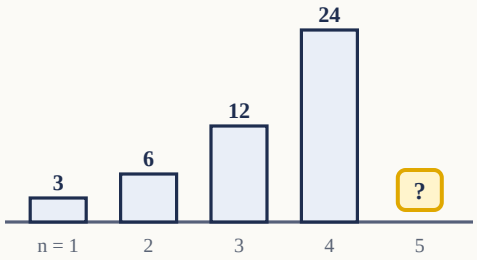


d = _____

answer: −4

3. *Next term*

What is the next term? 3, 6, 12, 24, …

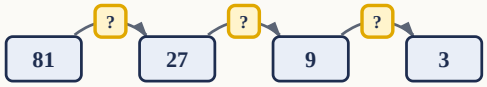


Answer: _____

answer: 48

4. *Common ratio*

What is the common ratio *r*? 81, 27, 9, 3, …



r = _____

answer: $\frac{1}{3}$

5. *Linear or exponential?*

Is this table linear or exponential?

x	0	1	2	3
y	2	6	18	54

Answer: _____

answer: exponential (×3 each time)

6. *Evaluate*

$f(x) = 3 \cdot 2^x$. Find f(3).

x	0	1	2	3
f(x)	3	?	?	?

f(3) = _____

answer: 24

SCORE 4.0 Target Content

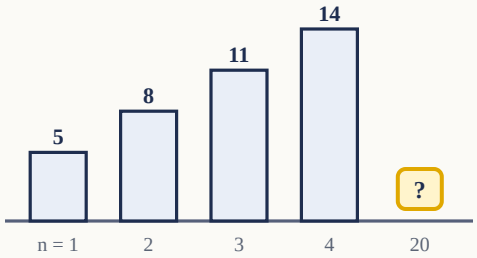
NC.M1.F-IF.3 · NC.M1.F-IF.8b · NC.M1.F-BF.1a · NC.M1.F-BF.2 · NC.M1.F-LE.3 Use sequence formulas and build exponential models.

• *I can use explicit and recursive rules, write growth models, find percent change, and compare linear and exponential growth.*

Solve each problem. Show your work on paper.

7. *Explicit formula*

An arithmetic sequence has $a_1 = 5$ and $d = 3$. Find a_{20}.



a_{20} =

answer: 62

8. *Recursive rule*

$a_1 = 10$ and $a_n = a_{n-1} - 4$. Find a_4.

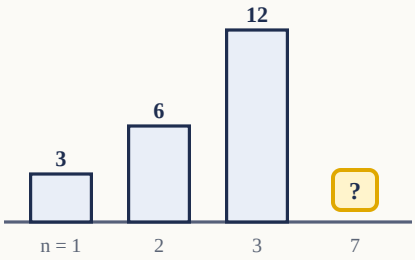


a_4 =

answer: −2

9. *Geometric term*

A geometric sequence has $a_1 = 3$ and $r = 2$. Find a_7.



a_7 =

answer: 192

10. *Write a model*

200 bacteria double every hour. Write P(t), the number after t hours.

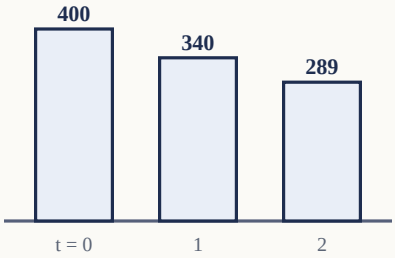
t (hours)	0	1	2
P	200	400	800

P(t) =

answer: $P(t) = 200(2)^t$

11. *Growth or decay*

$y = 400(0.85)^t$. By what **percent** does y decrease each year?

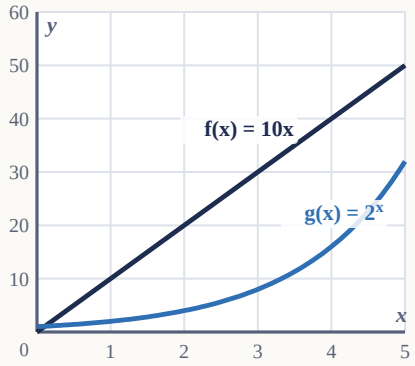


percent =

answer: 15%

12. *Compare growth*

$f(x) = 10x$ and $g(x) = 2^x$ are graphed for $0 \le x \le 5$. Which is greater when x = 10: f or g?



Answer:

answer: g (1024 > 100)

SCORE 5.0 Advanced Content

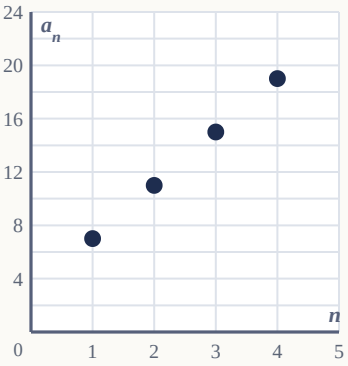
NC.M1.F-IF.3 · NC.M1.F-IF.7 · NC.M1.F-BF.1a · NC.M1.F-BF.2 · NC.M1.F-LE.1 · NC.M1.F-LE.5 Write formulas and interpret exponential models.

• *I can write explicit formulas, model growth and decay, explain the numbers in a model, and describe end behavior.*

Solve each problem. Some answers are words.

13. *Explicit formula*

Write an explicit formula for a_n: 7, 11, 15, 19, …

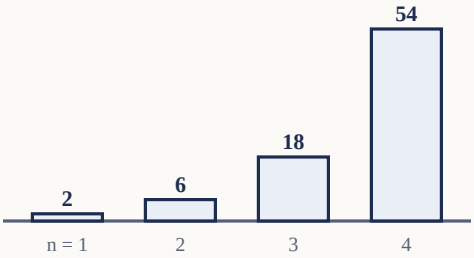


a_n =

answer: $a_n = 4n + 3$ (or $7 + 4(n - 1)$)

14. *Explicit formula*

Write an explicit formula for a_n: 2, 6, 18, 54, …

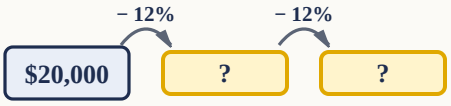


a_n =

answer: $a_n = 2(3)^{n-1}$

15. *Write a model*

A car is worth $20,000. It loses 12% of its value each year. Write V(t), its value after t years.



V(t) =

answer: $V(t) = 20000(0.88)^t$

16. *Interpret*

$P(t) = 1500(1.04)^t$ models a town's population. What is the yearly growth rate as a percent?

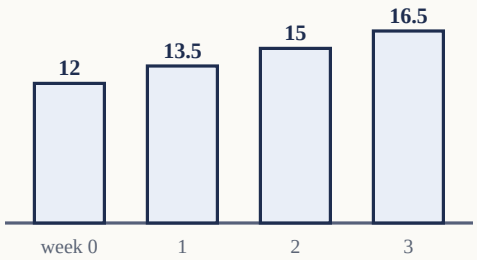


percent =

answer: 4%

17. *Linear or exponential?*

A plant is 12 cm tall. It grows 1.5 cm each week. Is its height linear or exponential?

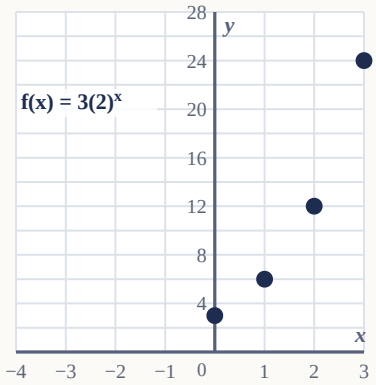


Answer:

answer: linear (adds 1.5 each week)

18. *End behavior*

$f(x) = 3(2)^x$. As x gets very negative ($x \to -\infty$), f(x) gets close to what number?



Answer:

answer: 0